# MA51900 Homework 1
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## Handout 1, Question 4 (David)

Six cookies are distributed completely at random, one by one, to six kids. What is the probability that Two kids get two cookies each, and another two get one each.

### Answer

Our sample space is defined by the number of ways six cookies can be distributed to the children. Since there are six cookies (ie six choices to make), and six possible recipients at each choice, the total count of our sample space is $|\Omega| = 6^6$.

Now we define the count of our event. The first four cookies will go to kids that don't yet have a cookie, then the last two will double up among the first four that already received a cookie.

$$A = \binom{6}{4}\binom{4}{2} = \frac{6!}{4!2!}\frac{4!}{2!2!} = \frac{5 \cdot 6}{2}\frac{3 \cdot 4}{2} = 15 \cdot 6 = 90$$

Therefore, we have

$$P(A) = \frac{90}{6^6} = \frac{15}{6^5}$$

## Handout 1, Question 8 (Andre)

If $n$ balls are placed completely at random into $n$ cells, find the probability that exactly one cell remains empty. Next, find the limit of this probability as $n$ tends to infinity rigorously.

### Answer

The probability that exactly one cell remains empty if $n$ balls are placed completely at random into $n$ cells is equivalent to one cell receiving 2 balls and $n-2$ cells receiving 1 ball each. That probability is

$$\frac{\binom{n}{1}\binom{n}{2}\binom{n-1}{n-2}(n-2)!}{n^n}$$

Which simplifies to

$$\frac{n(n-1)n!}{2n^n}$$

The limit of this probability as $n$ tends to infinity is evaluated using Stirling's approximation.

$$\lim_{n\to\infty} \frac{n(n-1)\sqrt{2\pi n}(\frac{n}{e})^n}{2n^n}$$

Simplifying

$$\frac{\sqrt{2\pi}}{2}(\lim_{n\to\infty} \frac{n^{5/2}}{e^n} - \lim_{n\to\infty} \frac{n^{3/2}}{e^n})$$

Evaluating the limits yields indefinite forms.

$$\frac{\sqrt{2\pi}}{2}(\frac{\lim_{n\to\infty} n^{5/2}}{\lim_{n\to\infty} e^n} - \frac{\lim_{n\to\infty} n^{3/2}}{\lim_{n\to\infty} e^n}) = \frac{\sqrt{2\pi}}{2}(\frac{\infty}{\infty} - \frac{\infty}{\infty})$$

Applying L'Hopital's rule.

$$\frac{\sqrt{2\pi}}{2}(\frac{5}{2}\lim_{n\to\infty} \frac{n^{3/2}}{e^n} - \frac{3}{2}\lim_{n\to\infty} \frac{n^{1/2}}{e^n})$$

The limits are still in indefinite form. Applying L'Hopital's rule again.

$$\frac{\sqrt{2\pi}}{2}(\frac{15}{4}\lim_{n\to\infty} \frac{n^{1/2}}{e^n} - \frac{3}{4}\lim_{n\to\infty} \frac{n^{-1/2}}{e^n})$$

Evaluating the second limit.

$$\begin{aligned}
\frac{3}{4}\lim_{n\to\infty} \frac{n^{-1/2}}{e^n} &= \frac{3}{4}\lim_{n\to\infty} \frac{1}{\sqrt{n}e^n} \\
&= \frac{3}{4}\frac{\lim_{n\to\infty} 1}{\lim_{n\to\infty} \sqrt{n}e^n} \\
&= \frac{3}{4} \cdot \frac{1}{\infty} \\
&= 0
\end{aligned}$$

The second limit is 0. The first limit is still in indefinite form. Applying L'Hopital's rule again.

$$\frac{\sqrt{2\pi}}{2}(\frac{15}{8}\lim_{n\to\infty} \frac{n^{-1/2}}{e^n})$$

Evaluating the limit.

$$\begin{aligned}\lim_{n\to\infty}\frac{n^{-1/2}}{e^n} &= \lim_{n\to\infty}\frac{1}{\sqrt{n}e^n}\\ &= \frac{\lim_{n\to\infty}1}{\lim_{n\to\infty}\sqrt{n}e^n}\\ &= \frac{1}{\infty}\\ &= 0\end{aligned}$$

Substituting the limits into the original equation.

$$\frac{\sqrt{2\pi}}{2}(\lim_{n\to\infty}\frac{n^{5/2}}{e^n}-\lim_{n\to\infty}\frac{n^{3/2}}{e^n})=\frac{\sqrt{2\pi}}{2}(0-0)=0$$

Therefore, the limit of this probability as $n$ tends to infinity is 0.

## Handout 1, Question 10 (David)

(Feller, Vol. I). A group of 100 boys and 100 girls are divided at random into two groups of 100 people. Find the probability that each group has an equal number of boys and girls. Replace 100 by $2N$ and derive a general formula.

### Answer

The sample space more than 100, because it must include different combinations of which boys and girls go into each group. Think of it like a coin toss per boy and girl. This is $2^{200}$. So, what is number of EXACTLY 100 H tosses out of 200 total tosses? $\binom{200}{100}$. That is the sample space.

$$\|\Omega\| = \binom{200}{100}$$

.

Our event $A$ is the event in which exactly 50 boys are in Group A. The count of this is $\binom{100}{50}$. However, we also must choose WHICH 50 girls end up in group A. This also ends up with a count of $\binom{100}{50}$. Therefore, the event count is $\binom{100}{50}^2$.

$$|A| = \binom{100}{50}^2$$

.

$$P(A)=\frac{\binom{100}{50}^2}{\binom{200}{100}}=\frac{(\frac{100!}{50!50!})^2}{\frac{200!}{100!100!}}=\frac{(100!)^4}{200!(50!)^4}=\frac{(51*52*...*100)^4}{200!}=0.1128$$

.

The probability is 11.28%.

We can generalize this as:

$$P(A) = \frac{\binom{2N}{N}^2}{\binom{4N}{2N}}$$

## Handout 1, Question 29 (Andre)

Six people are to be seated around a circular table.

### a)

What is the total number of ways that they can be seated?

#### Answer

If they were seated in a line, there would be 6! possible orderings.

$$6! = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 720$$

However, since they are sitting in a circular table, there are repeated orderings due to the invariance of rotational shifts. Due to this, we arbitrarily select the first person to sit anywhere, then find all the possible orderings relative to the first person.

$$1 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5! = 120$$

Therefore, there are 120 possible orderings to seat six people around a circular table.

### b)

What is the total number of ways that they can be seated if two people $X$ and $Y$ refuse to sit next to each other?

#### Answer

First arbitrarily seat X in a position. Then, seat each neighbor of X given the constraint.

$$1 \cdot 4 \cdot 3 = 12$$

Then we have three spots left with three possible people, one of which is $Y$.

$$3! = 3 \cdot 2 \cdot 1 = 6$$

Multiplying the possible neighbors of $X$ with the possible positions of $Y$.

$$12 \cdot 6 = 72$$

Therefore, there are 72 possible ways to seat 6 people around a circular table if two people $X$ and $Y$ refuse to sit next to each other.

## Handout 1, Question 31 (David)

The letters in INDIANA are randomly rearranged. What is the probability that on rearrangement, the letters will still read INDIANA?

### Answer

We treat each letter as unique for the sake of random combinations. The number of possible combinations for 7 letters is 7!.

$$\|\Omega\| = 7! = 5040$$

Event space must account for order AND for repeat letters. We have repeated letters for I, N, and A, with a total of two each. Therefore, these letters can switch while keeping the same word. The number of iterations in which INDIANA stays the same word, then, is found by $2 \cdot 2 \cdot 2$.

Note - When using ChatGPT to find mistakes, it corrected my initial approach in which I was using a method of 7 choose 2, 6 choose 2, 5 choose 1, etc. It showed how my sample space is much simpler, and can be counted with pairwise switching. Below is my derivation after the initial correction.

Therefore, we have:

$$\|A\| = 2 \cdot 2 \cdot 2 = 8$$

$$P(A) = \frac{2 \cdot 2 \cdot 2}{7!} = \frac{8}{5040} = \frac{1}{630} = 0.001587$$

The probability of randomly arranging the letters and still getting INDIANA is 0.1587%.

## Parzen, Page 5, Question 1.3 (Andre)

Describe how you would explain to a layman the meaning of the following statement:

> An insurance company is not gambling with its clients because it knows with sufficient accuracy what will happen to every thousand or ten thousand or a million people even when the company cannot tell what will happen to any individual among them.

### Answer

While an insurer may not know what will happen to a specific person they are able to price them a policy that will most likely make them profitable. They do this by comparing the individual to the history of the population. Using this they can find how likely it is that a person will be in an accident and price accordingly. Even if they are wrong for that specific person, if they have enough customers their predicted likelihood of accidents will be very accurate.

## Parzen, Page 16, Question 4.1 (David)

An experiment consists of drawing 3 radio tubes from a lot and testing them for some characteristic of interest. If a tube is defective, assign the letter $D$ to it. If a tube is good, assign the letter $G$ to it. A drawing is then described by a 3-tuple, each of whose components is either $D$ or $G$. For example, $(D, G, G)$ denotes the outcome that the first tube drawn was defective and the remaining 2 were good. Let $A_1$ denote the event that the first tube drawn was defective, $A_2$ denote the event that the second tube drawn was defective, and $A_3$ denote the event that the third tube drawn was defective. Write down the sample description space of the experiment and list all sample descriptions in the events

$$A_1,\ A_2,\ A_3,\ A_1 \cup A_2,\ A_1 \cup A_3,\ A_2 \cup A_3,\ A_1 \cup A_2 \cup A_3,\ A_1A_2,\ A_1A_3,\ A_2A_3,\ A_1A_2A_3.$$

### Answer

$$|\Omega| = 2^3 = 8$$

$$\Omega = (D, D, D), (D, D, G), (D, G, D), (D, G, G), (G, D, D), (G, D, G), (G, G, D), (G, G, G)$$

$$A_1 = (D, D, D), (D, D, G), (D, G, D), (D, G, G)$$

$$A_2 = (D, D, D), (D, D, G), (G, D, D), (G, D, G)$$

$$A_3 = (D, D, D), (D, G, D), (G, D, D), (G, G, D)$$

$$A_1 \cup A_2 = (D, D, D), (D, D, G), (D, G, D), (D, G, G), (G, D, D), (G, D, G)$$

$$A_1 \cup A_3 = (D, D, D), (D, D, G), (D, G, D), (D, G, G), (G, D, D), (G, G, D)$$

$$A_2 \cup A_3 = (D, D, D), (D, D, G), (D, G, D), (G, D, D), (G, D, G), (G, G, D)$$

$$A_1 \cup A_2 \cup A_3 = (D, D, D), (D, D, G), (D, G, D), (D, G, G), (G, D, D), (G, D, G), (G, G, D)$$

$$A_1A_2 = (D, D, D), (D, D, G)$$

$$A_1A_3 = (D, D, D), (D, G, D)$$

$$A_2A_3 = (D, D, D), (G, D, D)$$

$$A_1A_2A_3 = (D, D, D)$$

## Parzen, Page 22-23, Question 5.5 (Andre)

Let $A$ and $B$ be 2 events on a probability space. In terms of $P[A]$, $P[B]$, and $P[AB]$, express

### (i)

for $k = 0, 1, 2,\ P[\text{exactly } k \text{ of the events, } A \text{ and } B,\ \text{occur}]$,

### Answer

**k = 0**

$$\begin{aligned} P[\text{exactly 0 of the events, } A \text{ and } B, \text{ occur}] &= 1 - (P[A] + P[B] - P[AB]) \\ &= 1 - P[A] - P[B] + P[AB] \end{aligned}$$

**k = 1**

$$P[\text{exactly 1 of the events, } A \text{ and } B, \text{ occur}] = P[A] + P[B] - 2P[AB])$$

**k = 2**

$$P[\text{exactly 2 of the events, } A \text{ and } B, \text{ occur}] = P[AB])$$

### (ii)

for $k = 0, 1, 2,\ P[\text{at least } k \text{ of the events, } A \text{ and } B,\ \text{occur}]$,

**Answer**

**k = 0**

$$\begin{aligned} P[\text{at least 0 of the events, } A \text{ and } B, \text{ occur}] &= (1 - P[A] - P[B] + P[AB]) + (P[A] + P[B] - 2P[AB]) + (P[AB]) \\ &= 1 \end{aligned}$$

**k = 1**

$$P[\text{at least 1 of the events, } A \text{ and } B, \text{ occur}] = P[A] + P[B] - P[AB]$$

**k = 2**

$$P[\text{at least 2 of the events, } A \text{ and } B, \text{ occur}] = P[AB]$$

## (iii)

for $k = 0, 1, 2$, $P[\text{at most } k \text{ of the events, } A \text{ and } B, \text{ occur}]$,

**Answer**

**k = 0**

$$\begin{aligned} P[\text{at most 0 of the events, } A \text{ and } B, \text{ occur}] &= 1 - (P[A] + P[B] - P[AB]) \\ &= 1 - P[A] - P[B] + P[AB] \end{aligned}$$

**k = 1**

$$\begin{aligned} P[\text{at most 1 of the events, } A \text{ and } B, \text{ occur}] &= (1 - P[A] - P[B] + P[AB]) + (P[A] + P[B] - 2P[AB]) \\ &= 1 - P[AB] \end{aligned}$$

**k = 2**

$$\begin{aligned} P[\text{at most 2 of the events, } A \text{ and } B, \text{ occur}] &= (1 - P[AB]) + (P[AB]) \\ &= 1 \end{aligned}$$

## (iv)

$P[A \text{ occurs and } B \text{ does not occur}]$

**Answer**

$$P[A \text{ occurs and } B \text{ does not occur}] = P[A] - P[AB]$$

## Parzen, Page 22-23, Question 5.10 (David)

In a very hotly fought battle in a small war 270 men fought. Of these, 90 lost an eye, 90 lost an arm, and 90 lost a leg; 30 lost both an eye and an arm, 30 lost both and arm and a leg, and 30 lost both a leg and an eye; 10 lost all three. Find, for $k = 0, 1, 2, 3$, the number of men who suffered these injuries:

### (i)

exactly $k$

### (ii)

at least $k$

### (iii)

no more than $k$

### Answer

A = Lost an eye = 90

B = Lost an arm = 90

C = Lost a leg = 90

AB = Lost an eye and an arm = 30

AC = Lost an eye and a leg = 30

BC = Lost an arm and a leg = 30

ABC = Lost an arm, a leg, and an eye = 10

In these answers, we must account for double counts using Inclusion-Exclusion Formula

i, k=0: The number of soldiers that suffered no injury.

$Answer = \Omega - A - B - C + AB + AC + BC - ABC = 270 - 90 - 90 - 90 + 30 + 30 + 30 - 10 = 80$

i, k=1:

The number of soldiers that suffered exactly one injury.

$Answer = A+B+C-2AB-2AC-2BC+3ABC = 90+90+90-2*30-2*30-2*30+3*10 = 120$

(Citation - when checking my results, ChatGPT found a mistake in that I added ABC once instead of three times. This answer includes the correction.)

i, k=2:

The number of soldiers that suffered exactly two injuries.

$$Answer = AB + AC + BC - 3ABC = 30 + 30 + 30 - 3 * 10 = 60$$

i, k=3:

The number of soldiers that suffered exactly three injuries.

$$Answer = ABC = 10$$

ii, k=0:

The number of soldiers that suffered at least zero injuries. (Note - since all soldiers have, at a minimum, zero injuries, this includes the whole sample space.)

$$Answer = |\Omega| = 270$$

ii, k=1:

The number of soldiers that suffered at least 1 injury.

$$Answer = A + B + C - AB - AC - BC + ABC = 90 + 90 + 90 - 30 - 30 - 30 + 10 = 190$$

ii, k=2:

The number of soldiers that suffered at least 2 injuries.

$$Answer = AB + AC + BC - 2ABC = 30 + 30 + 30 - 2 * 10 = 70$$

ii, k=3: The number of soldiers that suffered at least 3 injuries.

$$Answer = ABC = 10$$

iii, k=0:

The number of soldiers that no more than 0 injuries. (Note - this is the same as suffered exactly 0 injuries from above.)

$$Answer = |\Omega| - A - B - C + AB + AC + BC - ABC = 270 - 90 - 90 - 90 + 30 + 30 + 30 - 10 = 80$$

iii, k=1: The number of soldiers with no more than one injury. (Note - this should equal the number of those with exactly 0 injuries plus those with exactly 1 injury.)

$$Answer = |\Omega| - AB - AC - BC + 2ABC = 270 - 30 - 30 - 30 + 20 = 200$$

iii, k=2:

The number of soldiers with no more than two injuries.

$$Answer = |\Omega| - ABC = 260$$

iii, k=3:

The number of soldiers with no more than three injuries. (Note - since all of the soldiers suffered only up to three injuries, this will include the whole sample space.)

$$Answer = |\Omega| = 270$$

## Parzen, Page 22-23, Question 5.11 (Andre)

Certain data obtained from a study of a group of 1000 subscribers to a certain magazine relating to their sex, marital status, and education were reported as follows: 312 males, 470 married, 525 college graduates, 42 male college graduates, 147 married college graduates, 86 married males, and 25 married male college graduates. Show that the numbers reported in the various groups are not consistent.

### Answer

For the sake of contradiction let us assume that the numbers reported are consistent. Then the size of any subset (or union of subsets) must be less than or equal to the size of the set ($N = 1000$). Therefore, the size of the union of the subsets male, married, and college graduate should be less than 1000.

**Married or College Graduate:** $M \cup C$

$$\begin{aligned} M \cup C &= M + C - M \cap C \\ &= 470 + 525 - 147 \\ &= 848 \end{aligned}$$

**Male or College Graduate:** $m \cup C$

$$\begin{aligned} m \cup C &= m + C - m \cap C \\ &= 312 + 525 - 42 \\ &= 795 \end{aligned}$$

**Male or Married:** $m \cup M$

$$\begin{aligned} m \cup M &= m + M - m \cap M \\ &= 312 + 470 - 86 \\ &= 696 \end{aligned}$$

**Male or Married or College Graduate:** $m \cup M \cup C$

$$\begin{aligned} m \cup M \cup C &= m + M + C - m \cap M - m \cap C - M \cap C + m \cap M \cap C \\ &= 312 + 470 + 525 - 42 - 147 - 86 + 25 \\ &= 1057 \end{aligned}$$

Since, the number of the union is larger than the sample size, the numbers reported are not consistent.